

Saif Ur Rehman

Machine learning engineer and AI researcher

Liverpool, United Kingdom saif10urrehman@gmail.com github.com/saifrhman linkedin.com/in/saifurrhmn
saifrhman.github.io

EDUCATION

- MSc Data Science and Artificial Intelligence**, University of Liverpool2025 – 2026
Dissertation: Beyond AlphaFold: Filling the Blind Spots in Protein Structure Prediction, supervised by Dr Olga Anosova.
- BE Electrical (Telecommunication) Engineering**, National University of Sciences and Technology (NUST), Pakistan2016 – 2021

RESEARCH

- Evidence and uncertainty in geometry foundation models**2026–present
Research in progress · Target: CVPR 2027 · First-author manuscript in development
When a feed-forward 3D/4D reconstruction model gives a confident answer, is that answer supported by the geometry in its input or by a learned prior that merely looks plausible? And could the model’s uncertainty tell the two apart?
- Peer-reward misalignment in multi-agent LLMs**2026
Under review · Submitted to a NeurIPS 2026 workshop
If two language-model agents grade each other and each is trained on the grade its peer assigns, is giving the solving and grading roles to different models enough to keep the training signal honest?
I wrote the single-agent RL training loop (REINFORCE with a moving-average baseline) that the two-agent trainer builds on, and produced the single-agent replication figure. In follow-up work I audited how policy-gradient credit reaches the grading tokens and designed the missing controls as a reward-topology comparison (objective reward, a frozen judge, a four-agent cycle, reciprocal pairs); it has not been run yet.
- Beyond AlphaFold: Filling the Blind Spots in Protein Structure Prediction**2026
MSc dissertation · OpenFold retraining in progress
Structure predictors such as AlphaFold2 and OpenFold are trained on the Protein Data Bank, where redundancy is usually controlled by sequence identity. What changes when redundancy is measured on backbone geometry instead, exactly and across the whole archive?
Sole author of the dissertation, supervised by Dr Olga Anosova (University of Liverpool). The invariant, the cleaning protocol and the multi-stage search strategy come from earlier BRI work on the PDB, and the OpenFold scripts started from a predecessor project. I re-implemented and scaled that protocol, implemented the compressed cover tree, and added the direct-edge removal policy, the provenance and validation layers, the sequence-redundancy comparison and the OpenFold data projection.
- Geographical Data and Computational Tools for Determining Restoration Priorities in Mangrove Ecosystems: A Scoping Review**2025–2026
Manuscript in preparation · Scoping review; began as MSc group coursework
Which remote-sensing data and machine-learning methods are used to decide where mangrove restoration is needed, how are they evaluated, and do they hold up outside the region they were developed in?
One of five co-authors, listed alphabetically by surname. I restructured the manuscript and drafted large parts of the sections on data sources, sensing modalities, computational methods and future work.

EXPERIENCE

- Applied AI Intern**, Rofsix2026 – present
Developing a real-time system that detects and redacts confidential information during screen sharing, combining OCR, transformer-based NER and image processing. Built and evaluated inference components with ONNX, OpenVINO and PaddleOCR, and profiled the pipeline for low-latency CPU execution.
- Senior Systems Engineer**, RoshTechNov 2022 – Sep 2025
Built threat-detection and anomaly-analysis workflows over MongoDB, Graylog and the Elastic Stack to support incident investigation for enterprise clients.
- Data Science Intern (part-time)**, Creative Tech SolutionsJan 2024 – Apr 2024
Python scripts for data analysis and data cleaning, supporting the analytics team.

Senior Cloud Engineer, Tier3tech

Feb 2022 – Nov 2022

Cloud security and Azure virtual desktop infrastructure; mentored team members.

Level 2 Support Team Lead, MiGo Innovations

Nov 2020 – Feb 2022

Led a team running network operations for offshore clients in the US.

SELECTED PROJECTS

F1 AI Copilot

github.com/saifrhman/f1-ai-copilot

A language model answering from memory can state Formula 1 article numbers, limits and penalties that are not in the regulations, with the same fluency as ones that are.

Hybrid receipt extraction

github.com/saifrhman/noise-robust-ocr-pipeline

Receipt datasets label a handful of fields, a useful schema needs many more, and a model trained on rule output inherits the rules' mistakes.

Football intelligence platform

github.com/saifrhman/football-analytics-platform

StatsBomb's open event data is deeply nested, match-scoped JSON; analysis needs typed, tested tables that can be traced back to the source.

TacticLens

github.com/saifrhman/tacticlens

Analysts looking for how an opponent behaves in situations like a given one search video by hand, using event tags that describe what happened rather than where the players were.

Closed-loop Bayesian optimisation

github.com/saifrhman/closed-loop-hpo-botorch

With a fixed budget of expensive evaluations, which configuration should be tried next?

COMPETITIONS

ICPC Northwestern Europe Regional Contest (NWERC) 2025: 107th, as the University of Liverpool team "Baby Harp Seal".

ICPC UK & Ireland Programming Contest 2025: 50th overall and 1st at the University of Liverpool site (team event).

Liverpool Computer Society Hackathon: 1st place (team).

Ultamation Hackathon: 2nd place, constraint-satisfaction scheduling.

SKILLS

Research and ML: PyTorch, RL fine-tuning of LLMs (LoRA, REINFORCE), Retrieval-augmented generation, Weak supervision, Bayesian optimisation (BoTorch, GPyTorch), Experimental design and error analysis

Geometry, vision and science: 3D/4D reconstruction models, Protein structure data (OpenFold, MMseqs2, gemmi, Mol*), Exact metric search (cover trees, k-d trees), Document AI (LayoutLMv3, EasyOCR), OpenCV

Systems and data: Python, C++, SQL, FastAPI, Streamlit, Qdrant, LangChain, BigQuery, dbt, Slurm and HPC clusters, Docker, GitHub Actions, Linux, ONNX, OpenVINO